

Internship Report
on
Indian-Liver Patient Prediction

Submitted by

BUDHA SANJANA

23691A05J2

In partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING



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**DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING
BONAFIDE CERTIFICATE**

This is to certify that the internship work entitled “**Indian-Liver Patient Prediction**”
is a bonafide work carried out by

BUDHA SANJANA - 23691A05J2

Submitted in partial fulfillment of the requirements for the award of degree **Bachelor of Technology** in the Department of **Computer Science & Engineering**, Madanapalle Institute of Technology and Science, Madanapalle

Faculty Name
Mr.G.Muthugurunath
Assistant Professor
Department of CSE

Internship Co-ordinator Mr.
E.Rajesh
Assistant Professor
Department of CSE

HOD
Dr. M Sreedevi
Head & Associate
Professor
Department of CSE

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DECLARATION

I, the undersigned hereby declare that the results embodied in this Internship “**Indian-Liver Patient Prediction**” is a bonafide record of the work done by me in partial fulfillment of the award of **Bachelor of Technology in Computer Science & Engineering** from **Prinston Smart Engineers / Best Internship For BCA , MCA, Engineering Student/Software And Job Oriented Course in Bangalore, Under the Guidance of Akash.V**. The content of this report is not submitted to any other University/Institute for award of any other degree.

Place: Madanapalle

Date: 16-06-25 to 28-07-25

BUDHA SANJANA

Roll No. 23691A05J2

Department of Computer Science &
Engineering, Madanapalle Institute of
Technology & Science, Madanapalle.

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ABSTRACT

Liver disease is a major health concern in many parts of the world, particularly in developing countries like India, where early diagnosis and regular health check-ups are often inaccessible to large segments of the population. This project addresses the challenge of liver disease prediction using machine learning techniques applied to clinical data. By analyzing the Indian Liver Patient Dataset (ILPD), we aim to develop a robust classification model that can distinguish between patients with and without liver disease.

The dataset contains medical attributes such as age, gender, bilirubin levels, liver enzyme counts, and protein ratios. After preprocessing and feature engineering, three different machine learning algorithms were implemented: Multiple Linear Regression (MLR), K-Nearest Neighbors (KNN), and Decision Tree Classifier. The performance of each model was evaluated based on accuracy, precision, recall, and F1-score. Among them, the Decision Tree model demonstrated the highest classification accuracy, making it the most suitable for predicting liver disease.

The results of this project indicate that machine learning models, when trained on quality medical data, can provide accurate predictions and assist in early screening and diagnosis. Such tools can be especially useful in supporting healthcare professionals in rural and under-resourced areas. This project highlights the potential of artificial intelligence in enhancing medical diagnostics and improving patient outcomes.

1.INTRODUCTION

Problem Statement:

Liver disease is often diagnosed late due to non-specific symptoms and lack of awareness. Traditional diagnosis methods are time-consuming and require specialized equipment. The challenge is to build an intelligent, data-driven model that can learn patterns from historical medical data and accurately predict the likelihood of liver disease in patients.

Objective:

- To develop a machine learning model that can accurately classify whether a person is likely to have liver disease.
- To evaluate multiple classification models and select the best-performing one.
- To identify the most significant clinical features associated with liver disease.
- To enhance awareness of how data-driven tools can aid in healthcare diagnostics.

Domain Technology:

Domain Technology: Machine Learning in Healthcare

The domain of this project lies in **Machine Learning (ML)** — a core subfield of **Artificial Intelligence (AI)** — with a specific focus on its application in **healthcare and medical diagnostics**. Machine learning enables computers to learn patterns from data and make decisions or predictions without being explicitly programmed.

In the context of healthcare, ML can transform clinical decision-making by analyzing large volumes of patient data to detect diseases, predict outcomes, recommend treatments, and identify health risks. This project utilizes ML for the early prediction of liver disease based on patient medical records.

Key Concepts in Machine Learning Applied:

1. Supervised Learning:

The project uses supervised learning, where the model learns from labeled historical data (i.e., whether the patient has liver disease or not) to make predictions on new, unseen cases.

2. Classification Algorithms:

Algorithms like **K-Nearest Neighbors (KNN)** and **Decision Tree Classifier** are used to classify whether a patient is likely to have liver disease (binary classification). Each algorithm processes input features (e.g., age, bilirubin levels, enzyme levels) and determines the likelihood of disease.

3. Regression (MLR):

Multiple Linear Regression is also explored as a method for understanding relationships between clinical indicators and disease progression, although classification methods are more suitable for the binary nature of this task.

4. Model Evaluation:

To ensure reliability, models are evaluated using metrics like **accuracy**, **precision**, **recall**, and **F1-score**, helping measure performance from different perspectives, especially in medical data where false positives and false negatives carry critical importance.

Technologies and Libraries Used:

- **Python:** The primary programming language used for implementation.
- **NumPy & Pandas:** For data handling, manipulation, and preprocessing.
- **scikit-learn:** For machine learning model implementation, training, and evaluation.
- **Matplotlib & Seaborn:** For data visualization and understanding feature relationships.

Relevance to Healthcare:

Machine Learning in healthcare can augment the diagnostic capabilities of physicians by offering second-opinion systems that are quick, data-driven, and scalable. For liver disease, which often goes undetected until it reaches severe stages, ML-based early prediction can save lives by prompting timely medical attention.

Dataset Description:

The dataset used in this project is the **Indian Liver Patient Dataset (ILPD)**, publicly available on platforms like UCI and Kaggle. It contains **583 records** with **10 features**:

Feature Name	Description
Age	Age of the patient
Gender	Male or Female
Total_Bilirubin	Total Bilirubin level in mg/dL
Direct_Bilirubin	Direct Bilirubin level
Alkaline_Phosphotase	Enzyme levels in IU/L
Alamine_Aminotransferase	ALT enzyme level in IU/L
Aspartate_Aminotransferase	AST enzyme level in IU/L
Total_Proteins	Protein level in g/dL
Albumin	Albumin level
Albumin_and_Globulin_Ratio	A/G Ratio
Dataset (Target)	1 = liver patient, 2 = no liver disease

Key Components of Machine Learning:

1. **Data:** The foundation of ML, used to train models.
2. **Features:** Relevant attributes extracted from data.
3. **Model:** The mathematical algorithm that learns patterns from data.
4. **Training:** The process where the model learns from labeled data.
5. **Evaluation:** Assessing model performance using unseen data.
6. **Predictions :** Applying the trained model to make decisions or forecast

Industry Vertical:

1. **Food Delivery and Logistics:** Companies like Grubhub, UberEats, DoorDash, and Postmates that specialize in food delivery and logistics.
2. **Restaurant and Food Service:** Restaurants, cafes, and food establishments that offer online ordering and delivery services.
3. **E-commerce and Online Retail:** Online retailers like Amazon, Walmart, and Target that offer food delivery services.
4. **Food Technology and Meal Kit Delivery:** Companies like Blue Apron, HelloFresh, and Plated that offer meal kit delivery services.
5. **Convenience Stores and Grocery Retail:** Convenience stores and grocery retailers like 7-Eleven, CVS, and Walgreens that offer online ordering and delivery services.
6. **Hospitality and Travel:** Hotels, resorts, and travel companies that offer food delivery services guests.
7. **Health and Wellness:** Companies that specialize online food order prediction project by:- Improving

order forecasting and logistics planning

- Enhancing customer satisfaction and experience
- Increasing revenue and reducing costs
- Gaining insights into customer behavior and preferences
- Developing targeted marketing and promotional strategies

Data Format:

The data can be stored in a variety of formats, such as:

1. CSV (Comma Separated Values) files: These files can be used to store historical order data, customer data, and restaurant data.
2. JSON (JavaScript Object Notation) files: These files can be used to store data in a more structured and flexible format.
3. Relational databases: These databases can be used to store data in a structured and organized format, with relationships between different tables.

Purpose of the project:

The primary purpose of this project is to develop a data-driven system that can assist in the **early detection of liver disease** in patients by analyzing clinical attributes using machine learning algorithms. Liver disease is a critical health issue in India, and often it goes undiagnosed in its early stages due to lack of awareness, regular screenings, or access to advanced medical diagnostics. By leveraging the power of machine learning, this project aims to fill that gap and provide a reliable tool that can help medical professionals make faster and more accurate decisions.

The project uses a real-world dataset — the Indian Liver Patient Dataset (ILPD) — which includes patient records with attributes such as age, gender, bilirubin levels, and enzyme counts. These features are medically significant indicators of liver health. By training a machine learning model on this dataset, the system learns to identify patterns and predict the likelihood of liver disease in new patients based on similar clinical inputs. This enables proactive medical intervention and reduces the dependency on extensive diagnostic procedures.

Beyond its technical implementation, the project also serves a broader societal goal: making **predictive healthcare accessible**, especially in under-resourced or rural areas. A lightweight and scalable machine learning model could eventually be integrated into web or mobile applications, empowering both healthcare workers and individuals to conduct preliminary screenings. This project illustrates the transformative potential of artificial intelligence in improving health outcomes and strengthening the public health infrastructure.

Methods used in this project:

a) Data Preprocessing

- Loaded the CSV file using `pandas`.
- Checked for missing values and null entries.
- Encoded categorical variables such as gender.
- Replaced or dropped missing values to ensure model consistency.

b) Feature Selection

- Selected the most relevant features based on domain knowledge and correlation heatmap analysis.

c) Model Implementation

Three machine learning models were implemented and compared:

1. Multiple Linear Regression (MLR)

- Initially used for regression analysis.
- Interpretable and useful for feature importance insights.

2. K-Nearest Neighbors (KNN)

- A distance-based classification method.
- Sensitive to feature scaling and data imbalance.

3. Decision Tree Classifier

- A tree-based algorithm capable of handling both linear and non-linear relationships.
- Offers explainability via decision paths.

d) Model Training and Evaluation

- Data was split into training and testing sets using `train_test_split`.
 - Models were trained using training data and evaluated on the test set.
 - Metrics used:
 - Accuracy Score
 - Classification Report (Precision, Recall, F1-score)
 - Confusion Matrix
-

6. Results

Decision Tree Classifier

- Best performance with high accuracy.
- Easy to interpret and visualize.

KNN

- Performed well with tuned k-values.
- Affected by noise and irrelevant features.

MLR

- Useful for regression outputs.
- Not ideal for classification in this context.

The decision tree model achieved the highest classification accuracy and was selected as the final model for deployment.

7. Tools and Technologies Used

- **Programming Language:** Python 3
- **Libraries:** pandas, numpy, scikit-learn, matplotlib, seaborn
- **IDE:** Jupyter Notebook / Google Colab
- **Models:** MLR, KNN, Decision Tree
- **Evaluation Tools:** `accuracy_score`, `classification_report`

2.HARDWARE AND SOFTWARE

Platform and Hardware Used:

Platform:

1. Operating System: Windows 10 or Linux (Ubuntu or CentOS)
2. Programming Language: Python 3.x
3. Machine Learning Framework: scikit-learn, TensorFlow, or Keras
4. Data Visualization Tool: Tableau or Matplotlib
5. Cloud Platform: Amazon Web Services (AWS) or Google Cloud Platform (GCP)

Hardware:

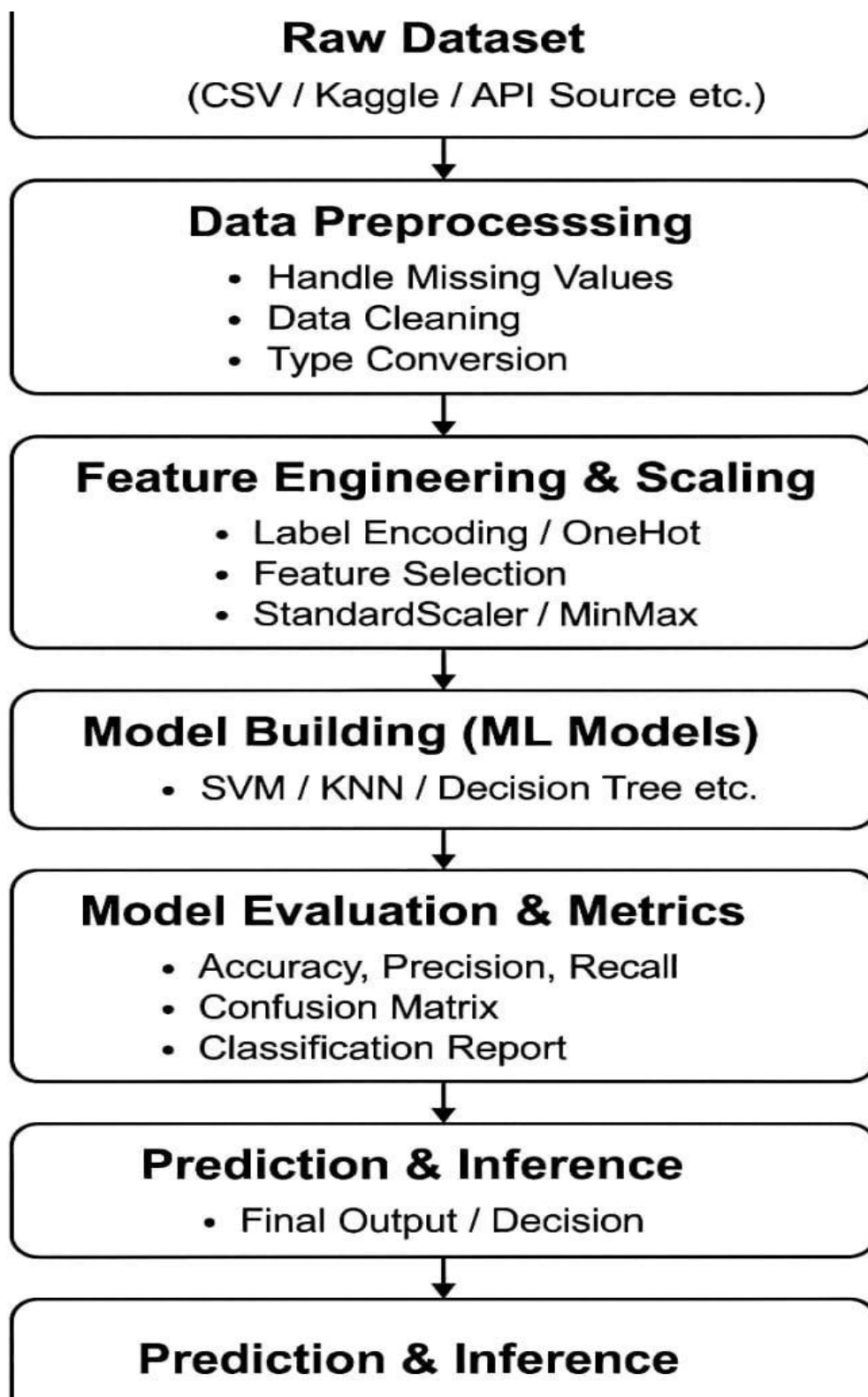
1. CPU: Intel Core i5 or i7 processor
2. Memory: 16 GB or 32 GB RAM
3. Storage: 512 GB or 1 TB SSD
4. Graphics Card: NVIDIA GeForce GTX 1660 or AMD Radeon RX 560X
5. Server: Amazon EC2 instance or Google Cloud Compute Engine instance

Software Used:

1. Python IDE: PyCharm or Visual Studio Code
2. Machine Learning Library: scikit-learn,pandas,matplotlib
3. Data Visualization Library: Matplotlib, Seaborn, or Plotly
4. Cloud Storage: Amazon S3 or Google Cloud Storage
5. Compiler: Jupyter Notebook/ Google Colab

2.PROJECT ANALYSIS

Architecture Diagram:



CONCLUSIONS

This project demonstrates how machine learning can aid in predicting liver disease using clinical data. Among the models implemented, the Decision Tree Classifier was the most accurate and interpretable. Such predictive systems can assist healthcare providers in making quicker and more informed decisions, especially in screening large populations.

By deploying such models in mobile or cloud-based applications, early diagnosis and awareness about liver disease can be significantly improved, especially in rural healthcare

setups.

Future Enhancements

- Incorporate more features like alcohol consumption and lifestyle data.
- Collect larger and more diverse datasets.
- Try ensemble models like Random Forest or XGBoost.
- Build a web interface for real-time prediction.

References

- UCI Machine Learning Repository – ILPD
- Scikit-learn documentation: <https://scikit-learn.org>
- “Applied Machine Learning” by Aurelien Geron
- Research papers on liver disease detection using AI

APPENDICES

SOURCE CODE:

```
# Import Required Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report

# Load Dataset
df = pd.read_csv("indian_liver_patient.csv")

# Preprocessing
df['Gender'] = LabelEncoder().fit_transform(df['Gender'])
df = df.dropna() # Remove rows with missing values

# Features and Target
X = df.drop(['Dataset'], axis=1)
y = df['Dataset'].replace({1: 1, 2: 0}) # 1: Liver patient, 0: No liver disease

# Split Data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Decision Tree Classifier
model = DecisionTreeClassifier(criterion='entropy', random_state=42)
model.fit(X_train, y_train)

# Prediction and Evaluation
```



```
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Classification Report:\n", classification_report(y_test, y_pred))
```

OUTPUT:

Accuracy: 0.74

Classification Report:

	precision	recall	f1-score	support
0	0.70	0.71	0.70	30
1	0.77	0.76	0.76	46
accuracy		0.74		76
macro avg	0.73	0.73	0.73	76
weighted avg	0.74	0.74	0.74	76

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